Whole numbers and yes-or-no choices

Optimization · Level 2 · Lesson 3



Today we will

- Say which decisions must be whole numbers or yes-or-no.
- Show how rounding a fraction can break a limit or cost more.
- Find the cheapest whole-number plan for our trip.

Remember

- Buses, beds and teachers come whole; loads are yes or no.
- Rounding down can break a limit; rounding up can cost more than the best.
- Points and priorities are human judgements: say whose they are.

Buses for the trip

1. Warm-up (5 min): 64 ÷ 30 = 2.13: what now?
2. Teach (8 min): Whole numbers, yes or no, rounding.
3. Table (15 min): Groups fill in seats and costs.
4. Compare (7 min): The best plan against rounding.
5. Fair? (5 min): Who rides the minibus?

Prepare one blank table per group. Use your own trip's numbers if you have them: the rounding problem appears with almost any numbers.

Let’s discuss

1. Why was the best whole-number plan not near either rounding?
2. Which decisions at school are yes or no?
3. Who should decide the points when we pack supplies for another school?

Exit ticket

- Give one decision that must be a whole number, and say what rounding could get wrong.